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COMPOSITE SYSTEM

$$\Psi^{(AB)} = \Psi^{(A)} \otimes \Psi^{(B)}$$

VECTORS

$$|a, b\rangle = |a\rangle \otimes |b\rangle$$

$$\forall |a\rangle \in \mathcal{H}^{(A)}, \forall |b\rangle \in \mathcal{H}^{(B)}$$

OPERATOR

$$F \otimes G | \psi, \phi \rangle = (F| \psi \rangle) \otimes (G| \phi \rangle)$$

$$F \otimes \mathbb{1} | \psi, \phi \rangle = (F| \psi \rangle) \otimes | \phi \rangle$$

States of composite system are still given by **NORMALIZED** VECTORS

$$| \psi^{(AB)} \rangle = | \psi^{(A)} \rangle \otimes | \phi^{(B)} \rangle \quad \text{SEPARABLE STATES (PRODUCT STATE)}$$

$$| \eta^{(AB)} \rangle = \frac{1}{\sqrt{2}} [| \psi_1^{(A)} \rangle \otimes | \phi_1^{(B)} \rangle + | \psi_2^{(A)} \rangle \otimes | \phi_2^{(B)} \rangle] \quad \text{ENTANGLED STATE}$$

→ NOT GUARANTEE THE ENTANGLEMENT

HOW DO WE GENERATE TWO OR MORE ENTAILED STATES

ENTANGLEMENT IS GENERALLY THE RESULT OF DYNAMICAL INTERACTION

NOT-INTERACTING SYSTEM: IF $\Psi^{(AB)} = \Psi^{(A)} + \Psi^{(B)}$

SHOW THAT $e^{A \otimes \mathbb{1}} = e^A \otimes \mathbb{1}$

$$| \psi^{(AB)} \rangle = | \psi^{(A)} \rangle \otimes | \phi^{(B)} \rangle \quad e^{\Psi^{(AB)}} \rightarrow e^A | \psi^{(A)} \rangle \otimes | \phi^{(B)} \rangle = e^A \otimes \mathbb{1} | \psi^{(AB)} \rangle$$

$$e^{A \otimes \mathbb{1}} | \psi^{(AB)} \rangle = \sum_n \frac{(A \otimes \mathbb{1})^n}{n!} | \psi^{(AB)} \rangle = \left(\sum_n \frac{A^n | \psi^{(A)} \rangle}{n!} \right) \otimes | \phi^{(B)} \rangle$$

$$(A \otimes \mathbb{1})^2 | \psi \rangle = (A \otimes \mathbb{1}) (A \otimes \mathbb{1}) | \psi \rangle = (A \otimes \mathbb{1}) [A | \psi^{(A)} \rangle \otimes | \phi^{(B)} \rangle] = A^2 | \psi^{(A)} \rangle \otimes | \phi^{(B)} \rangle$$

$$(A \otimes \mathbb{1})^n | \psi \rangle = A^n | \psi^{(A)} \rangle \otimes | \phi^{(B)} \rangle$$

ES. 6.14 A, B NOT INTERACTING $\Rightarrow e^{-\frac{i}{\hbar} H^{(AB)}} \epsilon = e^{-\frac{i}{\hbar} H^{(A)}} \epsilon \otimes e^{-\frac{i}{\hbar} H^{(B)}} \epsilon$

6.15 SUPPOSE THAT THE EVOLUTION OF $\overbrace{AB}^{\text{COMPOSITE SYSTEM}}$ IS GIVEN BY

$$U^{(AB)} = U^{(A)} \otimes U^{(B)}$$

IF THE SYSTEM IS IN A PRODUCT STATE, THEN IT REMAINS IN A PRODUCT STATE

$$\begin{aligned} | \psi^{(AB)}(\epsilon) \rangle &= U^{(AB)}(\epsilon) | \psi^{(AB)}(0) \rangle & | \psi^{(AB)}(0) \rangle &= | \psi^{(A)}(0) \rangle \otimes | \phi^{(B)}(0) \rangle \\ &= [U^{(A)} \otimes U^{(B)}] [| \psi^{(A)}(0) \rangle \otimes | \phi^{(B)}(0) \rangle] \end{aligned}$$

$$= U^{(A)} | \psi^{(A)}(0) \rangle \otimes U^{(B)} | \psi^{(B)}(0) \rangle = | \psi^{(A)}(t) \rangle \otimes | \psi^{(B)}(t) \rangle$$

$\underbrace{\quad}_{|\psi^{(A)}\rangle} \otimes \underbrace{\quad}_{|\psi^{(B)}\rangle}$

$$H^{(AB)} = \epsilon Z \otimes Z$$

$\nearrow$ PAULI MATRICES
 $\searrow$ STRUCTURE OF THE INTERACTION

$$H^{(AB)} = \epsilon Z \otimes Z$$

$$Z |z_+\rangle = |z_+\rangle$$

$$Z |z_-\rangle = -|z_-\rangle$$

$$\bullet H^{(AB)} |z_+, z_+\rangle = \epsilon Z \otimes Z = \epsilon (Z |z_+\rangle) \otimes (Z |z_+\rangle) = \epsilon |z_+\rangle \otimes |z_+\rangle = \epsilon |z_+, z_+\rangle$$

$$\bullet H^{(AB)} |z_-, z_-\rangle = \epsilon (Z |z_-\rangle) \otimes (Z |z_-\rangle) = \epsilon |z_-, z_-\rangle$$

$\downarrow$ $-|z_-\rangle$ $\downarrow$ $-|z_-\rangle$

$$\bullet H^{(AB)} |z_-, z_+\rangle = -\epsilon |z_-, z_+\rangle$$

$$\bullet H^{(AB)} |z_+, z_-\rangle = -\epsilon |z_+, z_-\rangle$$

$$A | \alpha \rangle = a | \alpha \rangle$$

$$e^A | \alpha \rangle = e^a | \alpha \rangle$$

$$U^{(AB)} \left(\epsilon = \frac{\pi \hbar}{4\epsilon} \right) = e^{i \frac{\pi}{4} Z \otimes Z}$$

$$U^{(AB)} |z_+, z_-\rangle = e^{-i \frac{\pi}{4} (+1)(-1)} |z_+, z_-\rangle$$

$$= e^{+i \frac{\pi}{4}} |z_+, z_-\rangle$$

$$U^{(AB)} |z_+, z_+\rangle = e^{-i \frac{\pi}{4}} |z_+, z_+\rangle$$

$$|x_+, x_+\rangle = |x_+^{(0)}\rangle \otimes |x_+^{(0)}\rangle$$

$$U^{(AB)}(t_2) = e^{-i\frac{\pi}{4}Z \otimes Z}$$

$$e^{-i\frac{\pi}{4}Z \otimes Z} |x_+, x_+\rangle$$

$$|x_+\rangle = \frac{1}{\sqrt{2}} (|z_+\rangle + |z_-\rangle)$$

$$\Rightarrow |x_+\rangle \otimes |x_+\rangle = \left[\frac{1}{\sqrt{2}} (|z_+\rangle + |z_-\rangle) \right] \otimes \left[\frac{1}{\sqrt{2}} (|z_+\rangle + |z_-\rangle) \right] = \frac{1}{2} [(|z_+\rangle + |z_-\rangle) \otimes (|z_+\rangle + |z_-\rangle)] = |x_+, x_+\rangle$$

$$e^{-i\frac{\pi}{4}Z \otimes Z} |x_+, x_+\rangle = \frac{1}{2} \left[e^{-i\frac{\pi}{4}} |z_+, z_+\rangle + e^{i\frac{\pi}{4}} |z_+, z_-\rangle + e^{i\frac{\pi}{4}} |z_-, z_+\rangle + e^{-i\frac{\pi}{4}} |z_-, z_-\rangle \right]$$

$$= \frac{1}{2} e^{-i\frac{\pi}{4}} \left[|z_+, z_+\rangle + e^{i\frac{\pi}{2}} |z_+, z_-\rangle + e^{i\frac{\pi}{2}} |z_-, z_+\rangle + |z_-, z_-\rangle \right]$$

$$= \frac{1}{2} e^{-i\frac{\pi}{4}} \left[|z_+\rangle \otimes [|z_+\rangle + i|z_-\rangle] + |z_-\rangle \otimes [|z_-\rangle + i|z_+\rangle] \right] =$$

$$= \frac{1}{2} e^{-i\frac{\pi}{4}} \left[|z_+\rangle \otimes [|z_+\rangle + i|z_-\rangle + i|z_-\rangle + |z_+\rangle] + |z_-\rangle \otimes [-i|z_-\rangle + |z_+\rangle] \right] =$$

$$= \frac{1}{\sqrt{2}} e^{-i\frac{\pi}{4}} \left[|z_+\rangle \otimes |y_+\rangle + i|z_-\rangle \otimes |y_-\rangle \right] \quad \text{ENTANGLED STATE}$$

OBSERVABLE AND COMPOSITE SYSTEM

NO COMMUNICATION THEOREM